AEROTHON 2026 - HAL x IITI

Problem Statement Title: **Physics-Informed Digital Twin for Real-Time Four-Stage Turbojet Health Monitoring**

Team Name: VYOMA

Participants: PREENITHI VARSHANA, JAYA MOUTHIKHAN, NARAYANAN ARUN, VAISHNAV ADITYA

Product Name: GARUDA

PROBLEM UNDERSTANDING & MOTIVATION

The Diagnostic Gap in Turbojet Health Monitoring

• ***As the PS states:*** *compressor fouling, turbine erosion, and combustor efficiency loss, **degrade** thrust, fuel efficiency, and reliability — but these health states **can't be measured directly**, only inferred from sensors.*

• ***High-fidelity simulations** capture this behavior well but are too **computationally expensive** for real-time use. The PS calls for a **surrogate model** instead — fast enough to **run continuously, still physically grounded**.*

• ***Gas-path parameter diagnostics** is already established in aero-engine prognostics research **(Rath, Mishra & Kushari, 2024)** — we extend it into exactly what's asked here: a **digital twin** reconstructing the **operational and health state** of this **single-spool, four-stage turbojet** from limited sensor data.*

• ***We tested this on the actual provided dataset** before assuming it: raw sensor readings alone explain **precisely "15.4%" of true health**, dominated by flight-condition noise. This is precisely why the **PS asks for a physics-informed approach, not a purely data-driven one**.*

What we're actually building:* a twin that shows its work, catches what a black box would miss**, *and answers the one question an engineer actually cares about — **can this engine fly today?

LITERATURE REVIEW / EXISTING APPROACHES

Three Generations of Approaches — and Where the Gap Actually Is

- *Classical:* ***Gas Path Analysis + Kalman Filter methods*** *physically model engine degradation, but generally* ***struggle under multi-fault, strongly nonlinear conditions*** *common in real deployments.*

- *Modern:* ***deep learning on C-MAPSS-style data*** *(LSTM, CNN, Transformers) gets high accuracy* ***but stays a black box*** *— no physical grounding, this is the default approach for this class of problem.*

- *2025 frontier:* ***physics-informed hybrids*** *(e.g. Pi-HT,* ***PINN-based*** *gas turbine models) fuse physics* ***constraints into the network*** *— still rare in practice, and rarely paired with real uncertainty quantification.*

- *Sensors don't just add noise — they can flat-out lie. A stuck, glitched, or garbage reading can look identical to a real fault, and* ***most health-monitoring approaches have no way to tell the difference. Ours does.***

PROPOSED METHODOLOGY & TECHNICAL APPROACH

Physics-Informed Hybrid Digital Twin — Two-Tier Architecture

• ***Pipeline:*** *Sensor Data → Physics Gas-Path Inversion →* ***Sensor Integrity Check*** *→* ***Health Estimation (two modes)*** *→ Unified Health Index &* ***Anomaly Detection*** *→ Decision Layer (RUL, Counterfactual, Recommendations) → Dashboard.*

• *Physics layer (no training required): Brayton-cycle station equations recover* ***compressor efficiency $\eta_c = (T2s - T01)/(T2 - T01)$*** *and* ***turbine efficiency $\eta_t = (T3 - T4)/(T3 - T4s)$*** *from ram-corrected stagnation conditions, plus* ***combustor pressure retention (P3/P2).***

• ***Two prediction modes****, chosen by data availability:* ***Flight-Log Match*** *(closed-form exact reconstruction, per-engine calibration) for known engines;* ***Fleet Baseline Estimate*** *(hierarchical trend + monotonicity-constrained gradient boosting + split conformal uncertainty)* ***for a brand-new engine with no history****.*

• ***Sensor Integrity Layer:*** *hard physical rules (verified 100% true on real data) plus a calibrated plausibility envelope, isolating "sensor is lying" from "****component is degrading****" — before either reaches the health estimate*

Key innovation: *physics does the interpretability, ML does the pattern extraction physics alone can't — fused, not stacked, and validated against a neural alternative we tested and rejected on its own merits (page 6).*

DATA SOURCES & PREPROCESSING

Official PS-2 Dataset — Structure, Split, and What We Found in It

• ***4 official files*** *(train.csv, test.csv, ground_truth.csv, complete reference set):* ***10 virtual engines, 30 recorded cycles each****, sensors matching the PS-2 schema exactly - (Altitude, Mach, RPM, Fuel Flow, P2/T2, P3/T3, P4/T4).*

• ***Split is per-cycle within each engine*** *(240 train / 60 test), not held-out engines — so validation uses* ***leave-one-engine-out*** *specifically to* ***test true generalization to an unseen engine****, not just an unseen cycle.*

• ***Preprocessing:*** ***physics inversion*** *runs first,* ***sensor integrity*** *check second,* ***ML features*** *third — raw sensors are never fed to a health model without passing through the physics layer.*

• ***Verified structural finding:*** *every engine's degradation follows an* ***identical normalized curve shape*** *(cross-engine deviation ≈* ***machine epsilon****) — this is what makes the* ***closed-form Flight-Log Match*** *mode possible, and it's a* ***property of the data we discovered****, not assumed.*

MODEL ARCHITECTURE / IMPLEMENTATION PLAN

Three-Tier Model Stack — Chosen by Testing, Not by Default

• ***Health estimation (Fleet Baseline mode):*** ***HistGradientBoostingRegressor*** *with a* ***monotonicity constraint on Cycle — health*** *is structurally forbidden from predicting an improvement over time, a physical law enforced at training time, not hoped for from data.*

• ***Anomaly & sensor-integrity detection:*** ***RandomForestRegressor*** *models the normal operating envelope per subsystem; deviations are checked against both a* ***statistical threshold*** *and* ***hard physical rules****.*

• ***Uncertainty:*** ***split conformal prediction*** *— a distribution-free coverage guarantee, not a bootstrap approximation.*

• ***Exact reconstruction (Flight-Log Match mode):*** ***closed-form alternating least squares*** *— no neural network. We tested a PCA + neural-network alternative and it scored* $R^2=-0.37$*; the* ***closed-form approach scored*** $R^2=1.0$*. With only 10 engines, there isn't enough data for a network to learn a useful prior — we picked the architecture the data justified, not the one that sounds more advanced.*

• ***Built entirely in Python/scikit-learn*** *— no deep learning framework needed, which is itself a validated decision, not a limitation.*

EXPECTED OUTCOMES & EVALUATION METRICS

Results Against the PS's Own Evaluation Criteria

•*Health Estimation Accuracy (30%):* ***R^2=1.0 for known engines*** *(exact closed-form reconstruction),* ***R^2=0.99 for held-out cycles*** *via the generalizable model — both on the official test split. On a fully* ***unseen engine*** *(leave-one-engine-out), 0.0113 RMSE.*

• *Surrogate Model Performance (20%):* ***Thrust R^2=0.99, TSFC R^2=0.98*** *— predicted directly from sensors in milliseconds, no iterative simulation required.*

• *Generalization Capability (15%):* **0.0113 RMSE** *via leave-one-engine-out testing — validated on an engine the model never saw during training.*

• *Physics Consistency (15%): discovered the exact composition law governing overall health (R^2=1.0),* ***monotonicity enforced structurally****, not just hoped for from data.*

• *Computational Efficiency (10%) + Dashboard & Interpretability (10%):* ***closed-form + lightweight gradient boosting*** *(no deep learning needed — validated, see Architecture), full interactive digital twin dashboard.*

Beyond the rubric:

→ ***Fault-injection test:*** *detection signal jumps from normal noise (z≈3) to z=20.65 on a simulated fault — instantly separable.*

→ ***Sensor Integrity Check:*** *zero false alarms across 300 real cycles, catches every injected garbage-sensor scenario tested.*

TEAM COMPOSITION & CONTRIBUTIONS

Team Composition

• ***Jaya Mouthikhan*** — ***Mechanical Engineering*** —
Built the physics layer:
Brayton-cycle station equations, compressor/turbine efficiency inversion, sensor plausibility rules.

• ***Narayanan Arun*** — ***Robotics and Artificial Intelligence*** —
Built the model architecture:
real-time state estimation for engine health, sensor fusion between the physics and ML layers, conformal uncertainty quantification, and the context-encoder ablation study.

• ***Preenithi Varshana*** — ***Cybersecurity*** —
Designed and ran the fault:
injection test suite and the sensor-integrity system: zero false alarms across all 300 real recorded cycles, and a fault signal that separates from normal noise by a 20x margin

• ***Vaishnav Aditya*** — ***Cybersecurity*** —
Built the interactive dashboard:
a live Three.js/WebGL 3D engine model, mission simulator, and real-time data pipeline, verified rendering with zero console errors.

APPENDIX & REFERENCES

References & Supplementary Material

References:
[1] Rath, N.; Mishra, R.K.; Kushari, A. "Aero engine health monitoring, diagnostics and prognostics for condition-based maintenance: An overview." Int'l J. Turbo & Jet-Engines, 2024, 40(s1), 279–292

[2] Kalman Filter Constraint Switching for Turbofan Engine Health Estimation, ScienceDirect

[3] Gas Path Health Monitoring for a Turbofan Engine Based on a Nonlinear Filtering Approach, MDPI Energies

[4] Physics-Informed Hybrid Transformer with Uncertainty Quantification for RUL of Turbofan Engines, 2025

[5] Physics-Informed Neural Networks for Industrial Gas Turbines: Recent Trends, Advancements and Challenges, 2025

Supplementary Material:
- *Live Interactive Dashboard: [AeroTwin link]*
- *Source Code & Technical Report: available on qualification for the Offline Final, per submission guidelines*
- *Official PS-2 Dataset: IIT Indore x HAL Google Drive repository*